Endpoint Derivative Analysis for Confounding Monotonicity

From Conjecture to Conditional Theorems

1 Setup and Notation

Let C_0 be a $K \times K$ column-stochastic matrix with diagonal entries $\alpha_j = (C_0)_{jj}$. Define $C(\delta) = (1-\delta)I + \delta C_0$ for $\delta \in [0,1]$ and

$$V_B(\delta) = \mathrm{Var}(\mathbb{E}[g(A) \mid A^*_\delta]) = \sum_j \frac{N_j(\delta)^2}{q_j(\delta)} - \bar{g}^2,$$

where $N_j(\delta) = \sum_k g_k C(\delta)_{jk} p_k$, $q_j(\delta) = \sum_k C(\delta)_{jk} p_k$, and $\bar{g} = \sum_k p_k g_k$.

At $\delta = 1$: $q_j \equiv q_j(1) = \sum_k (C_0)_{jk} p_k$, $\mu_j \equiv \mu_j(1) = \sum_k g_k (C_0)_{jk} p_k / q_j$.

Define the *estimation residual* $e_j = g_j - \mu_j(1)$.

2 Director's Formula for $V_B'(1)$

Proposition 1 (Endpoint derivative decomposition)**.**

$$V_B'(1) = \underbrace{\sum_j (p_j + q_j)\, e_j^2}_{\textit{LHS: residual energy}} \; \underbrace{- \sum_{j \neq k} (C_0)_{jk}\, p_k\, (g_j - g_k)^2}_{\textit{RHS: pairwise dispersion}}. \tag{1}$$

Proof. From the derivative formula established in Appendix B.3:

$$V_B'(\delta) = -\sum_{j,k} \left[(C_0)_{jk} - \mathbf{1}_{j=k}\right] p_k\, (g_k - \mu_j(\delta))^2. \tag{2}$$

At $\delta = 1$, split diagonal ($j = k$) from off-diagonal ($j \neq k$):

$$V_B'(1) = \sum_j (1 - \alpha_j)\, p_j\, e_j^2 - \sum_{j \neq k} (C_0)_{jk}\, p_k\, (g_k - \mu_j)^2.$$

Key identity. Since $\mu_j = \sum_k g_k (C_0)_{jk} p_k / q_j$:

$$e_j = g_j - \mu_j = \frac{1}{q_j} \sum_{k \neq j} (C_0)_{jk}\, p_k\, (g_j - g_k). \tag{3}$$

Substitute $g_k - \mu_j = -(g_j - g_k) + e_j$ into the off-diagonal term:

$$(g_k - \mu_j)^2 = (g_j - g_k)^2 - 2(g_j - g_k)\, e_j + e_j^2.$$

The off-diagonal sum becomes:

$$\begin{aligned}\sum_{j\neq k}(C_0)_{jk}\,p_k\,(g_k-\mu_j)^2 &= \sum_{j\neq k}(C_0)_{jk}\,p_k\,(g_j-g_k)^2 \\ &\quad -2\sum_j e_j \underbrace{\sum_{k\neq j}(C_0)_{jk}\,p_k\,(g_j-g_k)}_{=q_j e_j \text{ by (3)}} + \sum_j e_j^2 \underbrace{\sum_{k\neq j}(C_0)_{jk}\,p_k}_{=q_j-\alpha_j p_j} \\ &= \sum_{j\neq k}(C_0)_{jk}\,p_k\,(g_j-g_k)^2 - 2\sum_j q_j e_j^2 + \sum_j (q_j-\alpha_j p_j)\,e_j^2.\end{aligned}$$

Collecting the e_j^2 coefficients:

$$\begin{aligned}V_B'(1) &= \sum_j \big[(1-\alpha_j)p_j + 2q_j - (q_j-\alpha_j p_j)\big]\,e_j^2 - \sum_{j\neq k}(C_0)_{jk}\,p_k\,(g_j-g_k)^2 \\ &= \sum_j (p_j+q_j)\,e_j^2 - \sum_{j\neq k}(C_0)_{jk}\,p_k\,(g_j-g_k)^2. \qquad \square\end{aligned}$$

3 Quadratic Form Characterization

Proposition 2 (Matrix form)**.** *$V_B'(1) = \mathbf{g}^\top M\,\mathbf{g}$ where*

$$M = (I-T)^\top D_{p+q}\,(I-T) - (D_{p+q}-2W),$$

$W = C_0\,\mathrm{diag}(p)$, $T = D_q^{-1}W$ is row-stochastic, $D_q = \mathrm{diag}(q)$, $D_{p+q} = \mathrm{diag}(p+q)$.

Proof. The LHS of (1) equals $\mathbf{e}^\top D_{p+q}\,\mathbf{e}$ with $\mathbf{e} = (I-T)\mathbf{g}$. For the RHS, expand:

$$\begin{aligned}\sum_{j\neq k} W_{jk}(g_j-g_k)^2 &= \sum_j g_j^2(q_j - W_{jj}) - 2\sum_{j\neq k} W_{jk}g_j g_k + \sum_k g_k^2(p_k - W_{kk}) \\ &= \mathbf{g}^\top\big[D_{p+q} - 2W\big]\,\mathbf{g}. \qquad \square\end{aligned}$$

Remark 3. *$V_B'(1) \le 0$ for all $\mathbf{g}$ if and only if $M \preceq 0$ (negative semi-definite). This reduces the conjecture to a spectral condition on M.*

4 Theorem 1: Symmetric Doubly Stochastic C_0, Uniform p

Theorem 4 (Eigenvalue proof)**.** *Let C_0 be symmetric and doubly stochastic with $(C_0)_{jj} \ge \frac{1}{2}$ for all j, and let $p = \mathbf{1}/K$ (uniform). Then $V_B'(1) \le 0$ for all $\mathbf{g}$.*

Proof. Under uniform p: $q_j = \frac{1}{K}\sum_k (C_0)_{jk} = \frac{1}{K}$ (rows sum to 1), so $p_j + q_j = 2/K$ for all j.

From (1):

$$V_B'(1) = \frac{2}{K}\|\mathbf{e}\|^2 - \frac{1}{K}\sum_{j\neq k}(C_0)_{jk}(g_j-g_k)^2.$$

Step 1: Simplify $\|\mathbf{e}\|^2$. With $\mu = C_0\mathbf{g}$ and $\mathbf{e} = (I-C_0)\mathbf{g}$:

$$\|\mathbf{e}\|^2 = \mathbf{g}^\top(I-C_0)^2\mathbf{g} = \mathbf{g}^\top(I-2C_0+C_0^2)\mathbf{g}.$$

Step 2: Simplify the pairwise sum. Since C_0 is symmetric and doubly stochastic:

$$\begin{aligned}\sum_{j\neq k}(C_0)_{jk}(g_j-g_k)^2 &= \sum_{j,k}(C_0)_{jk}(g_j^2-2g_jg_k+g_k^2)\\ &= 2\|\mathbf{g}\|^2-2\mathbf{g}^\top C_0\mathbf{g}.\end{aligned}$$

Step 3: Combine.

$$\begin{aligned}V_B'(1) &= \frac{2}{K}\left[\|\mathbf{g}\|^2-2\mathbf{g}^\top C_0\mathbf{g}+\mathbf{g}^\top C_0^2\mathbf{g}\right]-\frac{1}{K}\left[2\|\mathbf{g}\|^2-2\mathbf{g}^\top C_0\mathbf{g}\right]\\ &= \frac{2}{K}\,\mathbf{g}^\top(C_0^2-C_0)\mathbf{g} = \frac{2}{K}\,\mathbf{g}^\top C_0(C_0-I)\mathbf{g}.\end{aligned}$$

Step 4: Eigenvalue analysis. Since C_0 is real symmetric, it has orthonormal eigenvectors $\{v_i\}$ with eigenvalues $\{\lambda_i\}$. The all-ones vector $\mathbf{1}$ is an eigenvector with $\lambda_1=1$. Expanding $\mathbf{g}=\sum_i c_iv_i$:

$$V_B'(1)=\frac{2}{K}\sum_i\lambda_i(\lambda_i-1)\,c_i^2.$$

Step 5: Sign of $\lambda_i(\lambda_i-1)$. By Gershgorin's theorem applied to row j: every eigenvalue lies in $[\alpha_j-(1-\alpha_j),\ \alpha_j+(1-\alpha_j)]=[2\alpha_j-1,\ 1]$. Since $\alpha_j\geq\frac{1}{2}$, all Gershgorin disks have left endpoint ≥ 0. Hence $\lambda_i\in[0,1]$ for all i, which gives $\lambda_i(\lambda_i-1)\leq 0$. Therefore $V_B'(1)\leq 0$. □

Corollary 5. *Under the conditions of Theorem 4, $V_B(\delta)$ is non-increasing on $[0,1]$ (since V_B is convex with $V_B'(1)\leq 0$).*

5 Theorem 2: Component-wise Diagonal Dominance

Theorem 6 (Sufficient condition for exact monotonicity)**.** *Let C_0 be column-stochastic with diagonal entries $\alpha_j=(C_0)_{jj}$. If*

$$\alpha_j\geq\frac{q_j}{p_j+q_j}\quad\text{for all } j \text{ with } p_j>0, \tag{4}$$

then $V_B'(1)\leq 0$ for all $\mathbf{g}$. Equivalently: $\alpha_jp_j\geq q_j(1-\alpha_j)$, i.e., the diagonal mass α_jp_j dominates the off-diagonal spillover $(1-\alpha_j)q_j$ in each row.

Proof. Define $\rho_j=\alpha_jp_j/q_j$ and weights $w_{jk}=(C_0)_{jk}p_k/q_j$ for $k\neq j$. Note $\sum_{k\neq j}w_{jk}=1-\rho_j$.

By Cauchy–Schwarz on (3):

$$e_j^2=\Big(\sum_{k\neq j}w_{jk}(g_j-g_k)\Big)^2\leq(1-\rho_j)\sum_{k\neq j}w_{jk}(g_j-g_k)^2=\frac{1-\rho_j}{q_j}\sum_{k\neq j}(C_0)_{jk}\,p_k\,(g_j-g_k)^2.$$

Hence:

$$(p_j+q_j)\,e_j^2\leq\frac{(p_j+q_j)(1-\rho_j)}{q_j}\cdot S_j,\qquad S_j=\sum_{k\neq j}(C_0)_{jk}\,p_k\,(g_j-g_k)^2.$$

From (1), $V_B'(1)\leq 0$ follows if $\frac{(p_j+q_j)(1-\rho_j)}{q_j}\leq 1$ for every j, which simplifies to:

$$(p_j+q_j)(q_j-\alpha_jp_j)\leq q_j^2\iff q_j(1-\alpha_j)\leq\alpha_jp_j\iff\alpha_j\geq\frac{q_j}{p_j+q_j}.$$ □

Remark 7 (Interpretation)**.** *Condition* (4) *is a* joint *condition on C_0 and p. It requires that the classifier's accuracy α_j for class j exceeds a threshold that depends on how much probability mass other classes "spill" into class j.*

For doubly stochastic C_0 with uniform p: $q_j = 1/K = p_j$, so condition (4) *reduces to $\alpha_j \geq 1/2$, recovering the sufficient condition in Theorem 4.*

Remark 8 (Tightness)**.** *The condition is tight for $K = 2$: the $K = 2$ proof in Appendix B.3 shows $V_B'(1) \leq 0$ iff $a + b \geq 1$ (i.e., $\alpha_j \geq 1 - \alpha_{1-j}$), while condition* (4) *requires $a \geq q_0/(p_0 + q_0)$ and $b \geq q_1/(p_1 + q_1)$. These are equivalent for $K = 2$, confirming tightness.*

6 Theorem 3: Quantitative Violation Bound

When condition (4) fails, we bound the magnitude of the positive part of $V_B'(1)$.

Theorem 9 (Violation bound)**.** *For column-stochastic C_0 with $\alpha_j = (C_0)_{jj} \geq c$ for all j, where $c \in (1/K, 1)$:*

$$V_B'(1) \leq (1 - c)^2 \, R_g^2, \tag{5}$$

where $R_g = \max_k g_k - \min_k g_k$ is the range of g.

Proof. Using the Cauchy–Schwarz-based per-component bound from Theorem 6:

$$V_B'(1) \leq \sum_j S_j \, \frac{p_j [q_j(1 - \alpha_j) - \alpha_j p_j]_+}{q_j^2},$$

where $[x]_+ = \max(x, 0)$ and $S_j = \sum_{k \neq j} (C_0)_{jk} \, p_k (g_j - g_k)^2$.

Since $S_j \leq (q_j - \alpha_j p_j) \, R_g^2$ and $q_j(1 - \alpha_j) - \alpha_j p_j \leq q_j(1 - c)$:

$$V_B'(1) \leq (1 - c) \, R_g^2 \sum_j \frac{p_j (q_j - c \, p_j)}{q_j} = (1 - c) \, R_g^2 \Big(1 - c \sum_j \frac{p_j^2}{q_j} \Big).$$

By Cauchy–Schwarz, $\sum_j p_j^2 / q_j \geq (\sum_j p_j)^2 / \sum_j q_j = 1$, so:

$$V_B'(1) \leq (1 - c)^2 \, R_g^2. \qquad \square$$

Remark 10 (Application to the paper's setting)**.** *For $g_k = \mathbb{E}[Y \mid A{=}k]$ with $|\beta| \leq 1$: $R_g \leq 2|\beta|_\infty \leq 2$, giving $V_B'(1) \leq 4(1 - c)^2$. With $c \geq 0.7$: $V_B'(1) \leq 0.36$. With $c \geq 0.9$: $V_B'(1) \leq 0.04$.*

7 From $V_B'(1)$ to Step Deviation ε

The conjecture bounds the *step deviation* in $|B(C(\delta))|$, not $V_B'(1)$ directly. The following proposition connects them.

Proposition 11 (Convex overshoot bound)**.** *Since $V_B(\delta)$ is convex with $V_B'(0) \leq 0$, when $V_B'(1) > 0$ there exists a unique $\delta^* \in (0, 1)$ with $V_B'(\delta^*) = 0$. The total non-monotone increase satisfies:*

$$V_B(1) - V_B(\delta^*) \leq \frac{[V_B'(1)]^2}{2 \, a}, \tag{6}$$

where $a = \inf_{\delta \in [0,1]} V_B''(\delta) > 0$ is the minimum curvature.

Proof. By convexity, V_B' is non-decreasing. Since $V_B'(\delta^*) = 0$ and $V_B'(1) > 0$:

$$V_B(1) - V_B(\delta^*) = \int_{\delta^*}^{1} V_B'(\delta)\, d\delta \leq V_B'(1)\,(1 - \delta^*).$$

Also, $V_B'(1) - V_B'(\delta^*) = V_B'(1) \leq a\,(1 - \delta^*)$ by convexity, so $1 - \delta^* \leq V_B'(1)/a$. □

Remark 12 (Numerical validation)**.** *Across 20,000 random configurations with $\alpha_j \geq 0.1$, the ratio of step deviation to $V_B'(1)$ satisfies $\varepsilon/V_B'(1) \leq \Delta\delta$ (the discretization step), with correlation > 0.999. This is because the non-monotone region is concentrated near $\delta = 1$, and within one discretization step the increase is approximately $V_B'(1) \cdot \Delta\delta$.*

8 Numerical Findings on $\alpha_j \geq 1/2$

Theorem 4 proves $V_B'(1) \leq 0$ for symmetric doubly stochastic C_0 with $\alpha_j \geq 1/2$ and uniform p. Theorem 6 extends this to doubly stochastic C_0 (not necessarily symmetric) with uniform p, since row sums equal 1 gives $q_j = 1/K = p_j$, making the DD condition $\alpha_j \geq 1/2$.

For *general* column-stochastic C_0 with $\alpha_j \geq 1/2$ and non-uniform p, $V_B'(1) \leq 0$ is **not** universally guaranteed. Numerical testing (100,000 random configurations with guaranteed $\alpha_j \geq 0.5$) finds:

- $V_B'(1) > 0$ in 0.69% of cases;
- All counterexamples have extreme class imbalance ($p_{\max}/p_{\min} > 300$);
- Maximum step deviation: $\varepsilon < 0.001$.

The failure mechanism: when p_j is very small for some class j, the spillover $q_j = \sum_k (C_0)_{jk} p_k$ can satisfy $q_j > p_j$ even with $\alpha_j \geq 1/2$, violating the DD condition $\alpha_j \geq q_j/(p_j + q_j)$.

9 Analysis of Failure Modes

9.1 When does condition (4) fail?

The condition $\alpha_j \geq q_j/(p_j + q_j)$ fails when q_j/p_j is large—i.e., class j receives disproportionate spillover from other classes relative to its own prior probability.

Lemma 13. *For doubly stochastic C_0, condition (4) requires:*

$$\left(\frac{\alpha_j}{1 - \alpha_j}\right)^2 \geq \frac{\max_k p_k}{\min_k p_k}.$$

Proof. For doubly stochastic C_0: $\sum_{k \neq j} (C_0)_{jk} = 1 - \alpha_j$, so $q_j = \alpha_j p_j + \sum_{k \neq j} (C_0)_{jk} p_k \leq \alpha_j p_j + (1 - \alpha_j) p_{\max}$. The condition $\alpha_j^2 p_j \geq (1 - \alpha_j) \sum_{k \neq j} (C_0)_{jk} p_k$ is most stressed at the smallest $p_j = p_{\min}$: $\alpha_j^2 p_{\min} \geq (1 - \alpha_j)^2 p_{\max}$. □

9.2 Near-permutation matrices

The adversarial tests identify near-permutation matrices as the hardest case. For $C_0 = (1 - \varepsilon) P_\sigma + \varepsilon/K \cdot \mathbf{1}\mathbf{1}^\top$ where P_σ is a permutation:

- $\alpha_j = \varepsilon/K$ if $\sigma(j) \neq j$ (most entries for derangements);

- $q_j \approx (1-\varepsilon)p_{\sigma^{-1}(j)} + \varepsilon/K$ can be large relative to p_j.

These matrices have $\alpha_j \ll 1/2$, causing condition (4) to fail. However, the permutation structure ensures that misclassification is *structured* (class j maps primarily to class $\sigma(j)$), which limits the actual bias violation.

10 Summary of Results

Condition on C_0, p	**Result**	**Method**
$K = 2$, $\alpha_j \geq 1/2$	$V_B'(1) \leq 0$ (exact)	Direct algebra
$K = 3$ symmetric, any p	$V_B'(1) \leq 0$ (exact)	Pointwise derivative
DS, $\alpha_j \geq 1/2$, uniform p	$V_B'(1) \leq 0$ (exact)	DD condition (Thm 6)
DS symmetric, $\alpha_j \geq 1/2$, uniform p	$V_B'(1) \leq 0$ (exact)	Eigenvalue (Thm 4)
$\alpha_j \geq q_j/(p_j + q_j)$ $\forall j$	$V_B'(1) \leq 0$ (exact)	Cauchy–Schwarz (Thm 6)
$\alpha_j \geq c > 1/K$, any p	$V_B'(1) \leq (1-c)^2 R_g^2$	Bound (Thm 9)
$\alpha_j \geq 1/2$, any p	$\varepsilon < 0.001$ (numerical)	10^5 random configs
General column-stochastic	$\varepsilon < 0.017$ (numerical)	545,945 adversarial curves

The gap between the provable sufficient conditions and the conjectured universal ε-bound lies in the interaction between the spillover structure of C_0 and the prior p: when p_j is very small for some class j, the spillover ratio q_j/p_j can exceed the threshold $\alpha_j/(1-\alpha_j)$, defeating per-component Cauchy–Schwarz. Yet the resulting violations are geometrically small because the convex overshoot (**??**) concentrates the non-monotone region in a narrow interval near $\delta = 1$.

A proof of the exact conjecture for general column-stochastic matrices likely requires exploiting global structure (e.g., Birkhoff decomposition, spectral properties of M) rather than per-row analysis.